

Utility-Scale BESS Value in a Synthetic Atlantic Electricity System

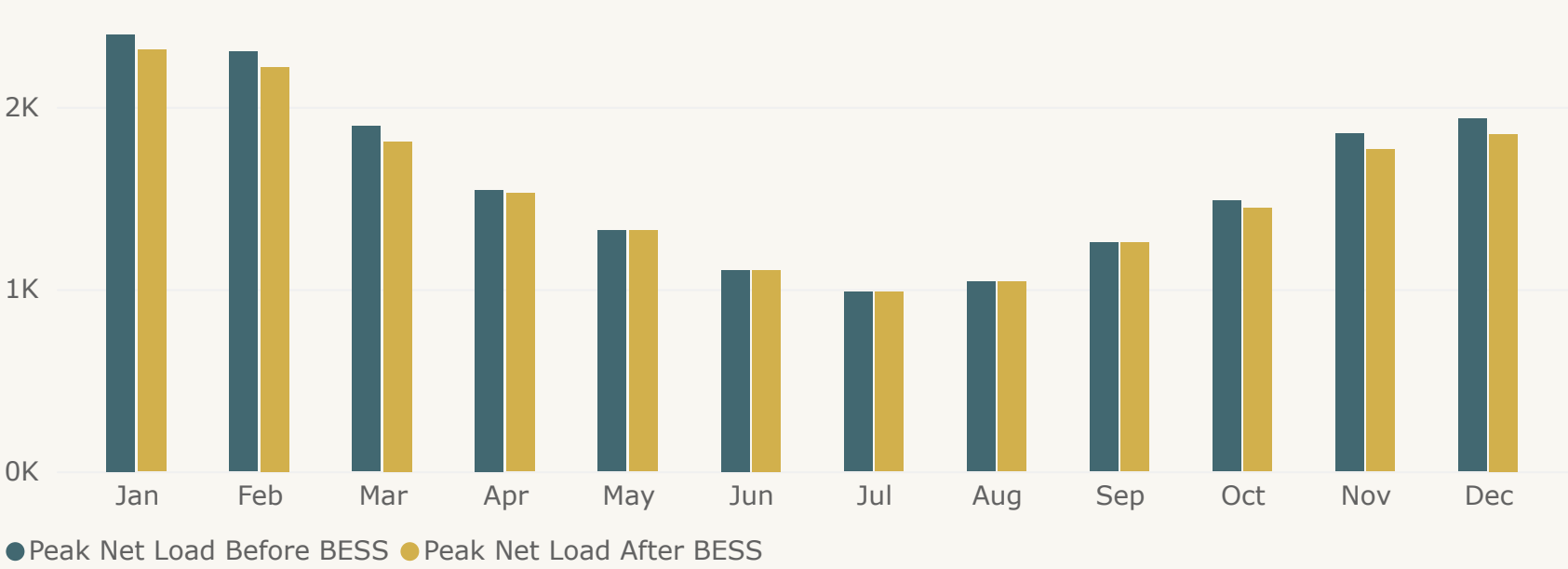
100 MW / 400MWh storage simulation using selective daily dispatch

0.8M	85	3.5%	34.0K	100	147
BESS Gross Avoided Cost (\$)	Peak Reduction (MW)	Peak Reduction (%)	BESS Energy Discharged (MWh)	BESS Equivalent Full Cycles	BESS Days Operated

BESS Gross Avoided Cost (\$)



Peak Net Load Before and After BESS (MW)



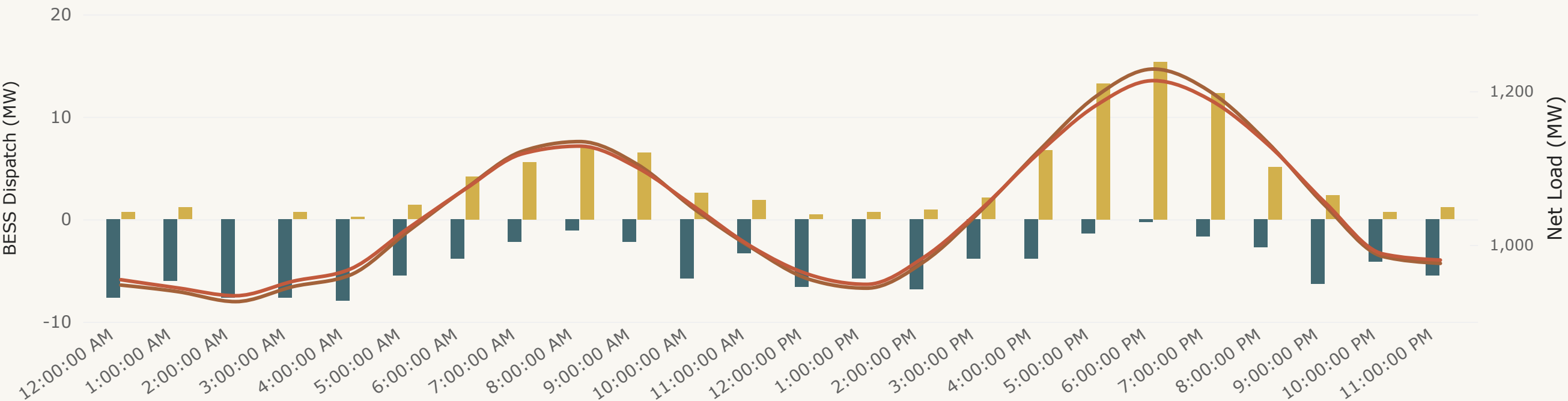
Finding: A 100 MW / 400 MWh BESS provides measurable value, but the opportunity is concentrated in winter. Under a selective daily dispatch rule, the battery produces approximately \$0.8 million in gross avoided cost, reduces the modeled annual peak by 85 MW, and completes approximately 100 equivalent cycles. Additional capacity, reliability, and ancillary-service

How the Battery Operates

Select Month for Hourly Profile

All

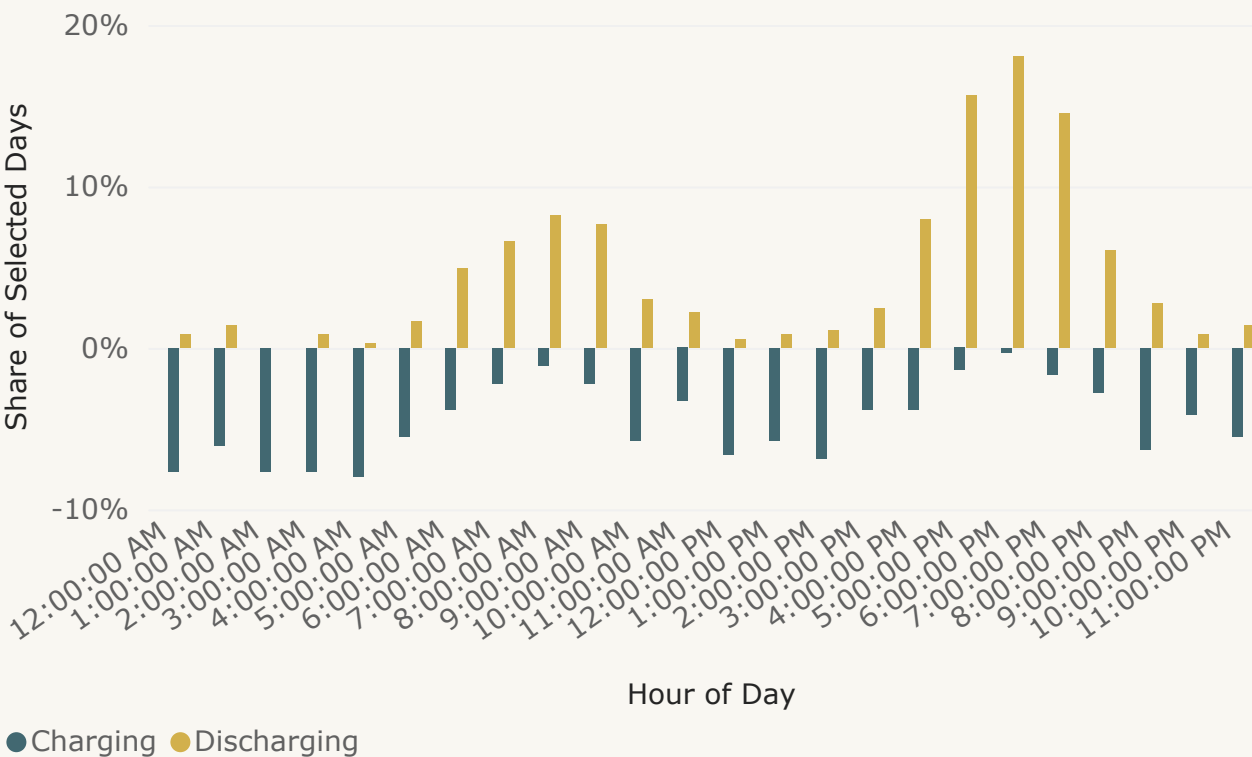
Average Hourly BESS Operation and Net Load: Selected Period



How it works: The modeled BESS charges during selected lower-cost hours and discharges during selected higher-cost hours. It shifts energy rather than creating it, returns approximately 85% of the energy used for charging, and reduces net load during targeted peak conditions.

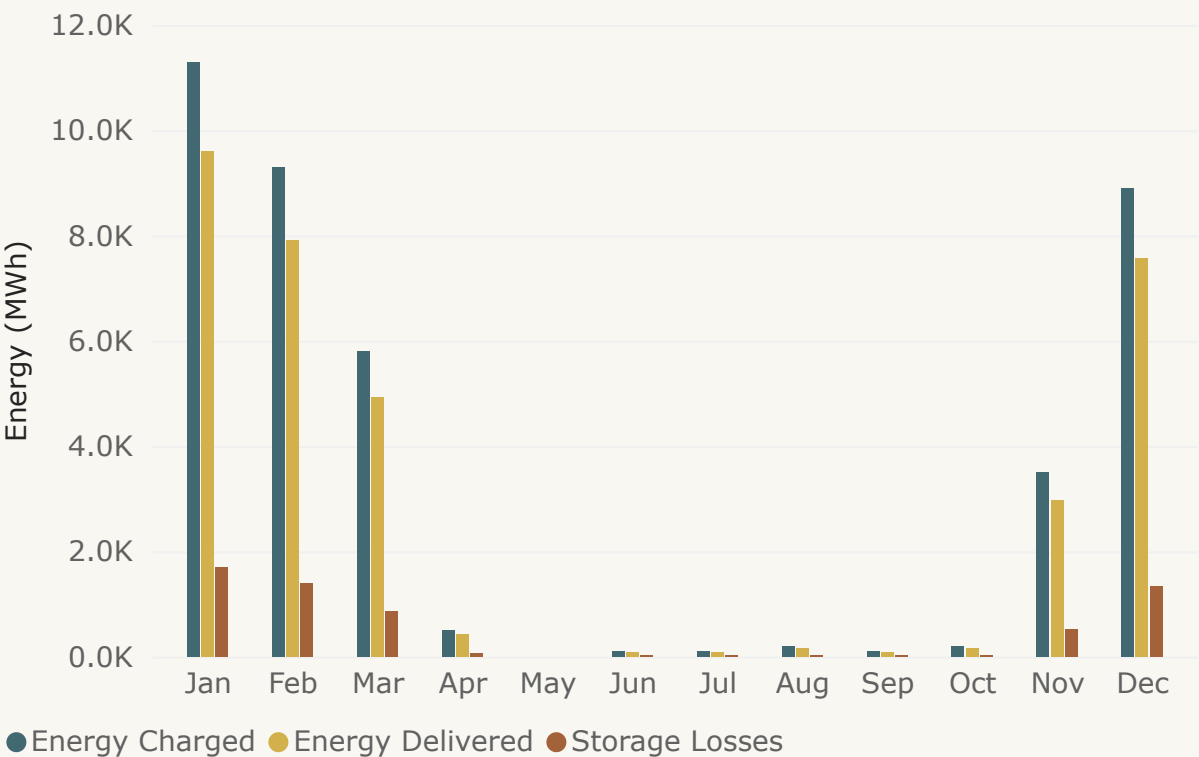
How Often the BESS Operates by Hour

Charging is shown below zero; discharging is shown above zero



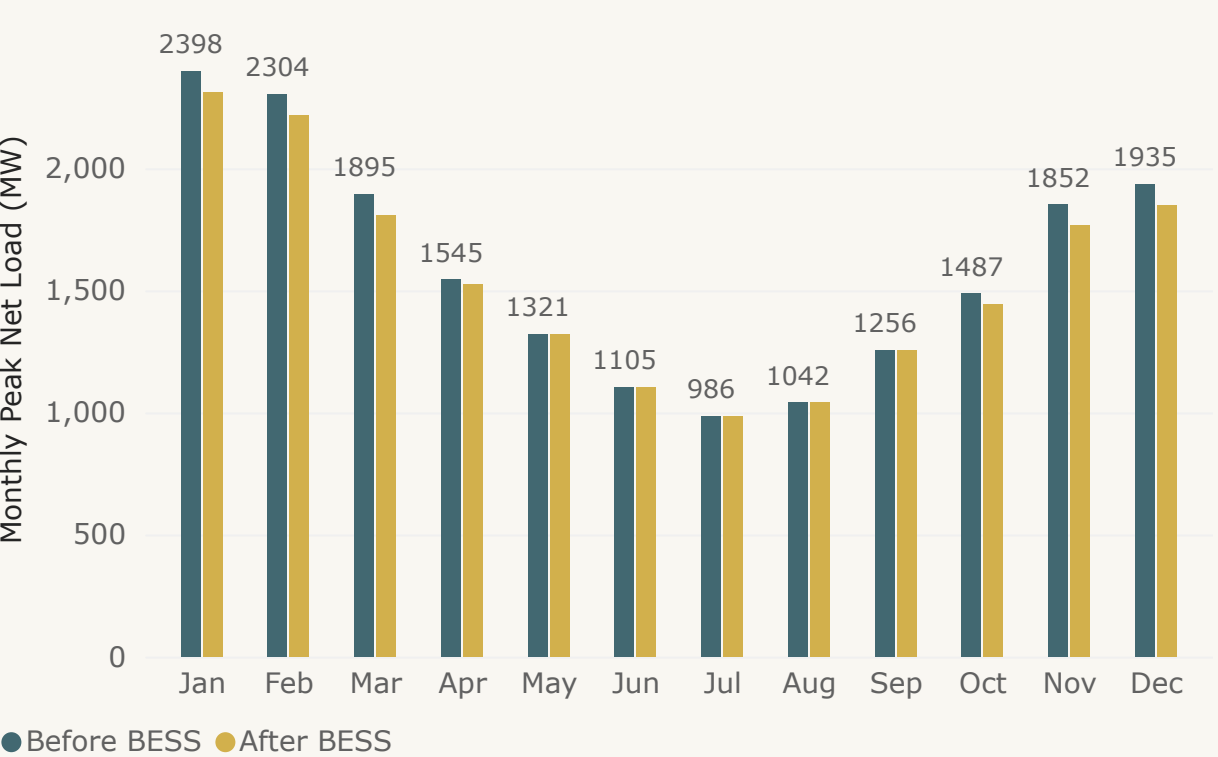
Monthly BESS Energy Balance

Energy delivered is lower than energy charged because of storage losses



Effect of BESS on Monthly Peak Net Load

Comparison of maximum hourly net load before and after storage dispatch



Peak Management and System Value

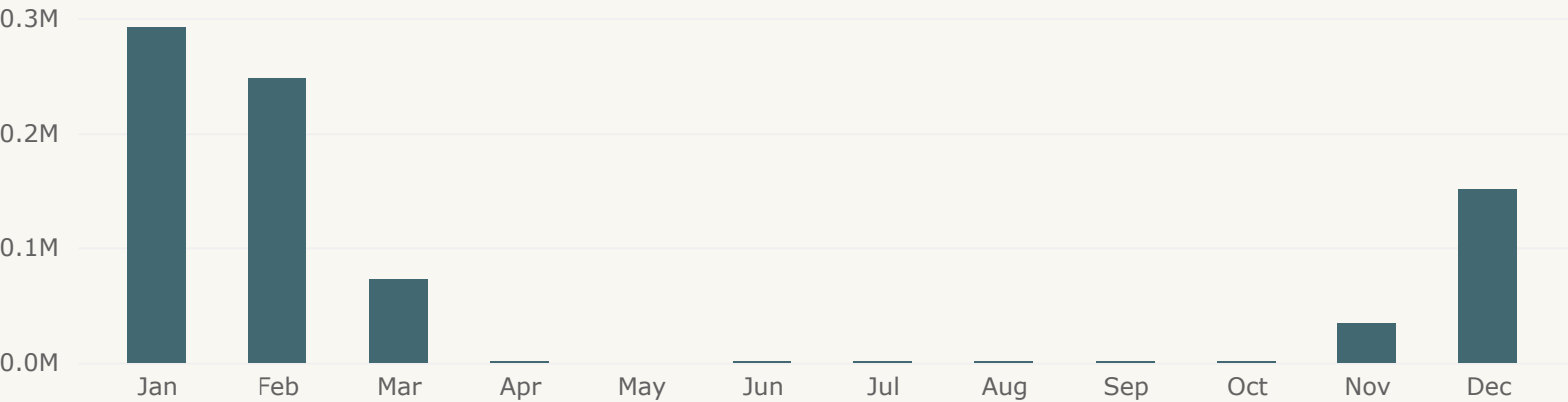
Gross Energy-Cost Value (\$)	Maximum Peak Reduction (MW)	Equivalent Full Cycles	Days with BESS Operation
0.80M	85	100	147

Select month for monthly profile

All

Monthly Gross Avoided System Cost

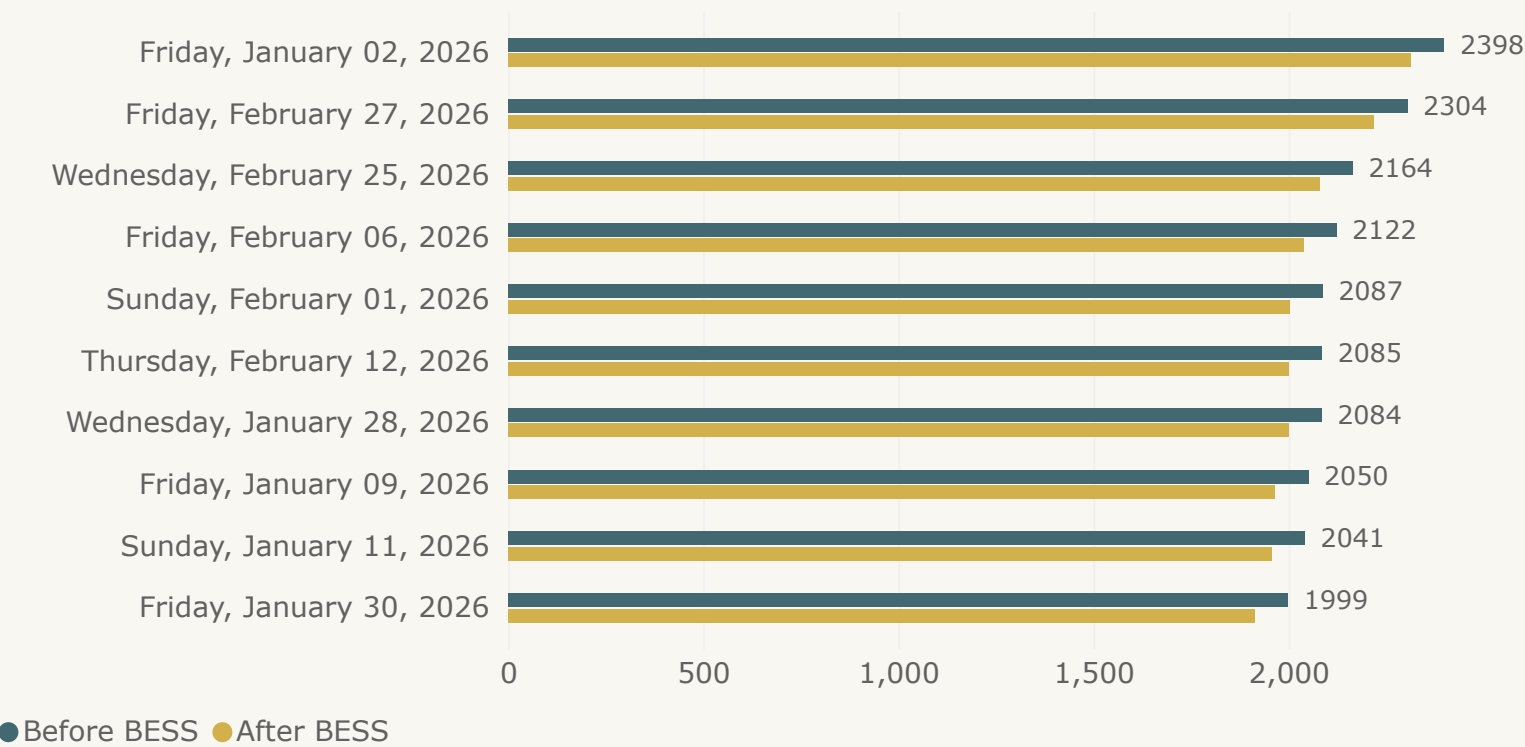
Value from selective charging and discharging based on hourly marginal cost



Share of Value Dec-Feb
86.3%

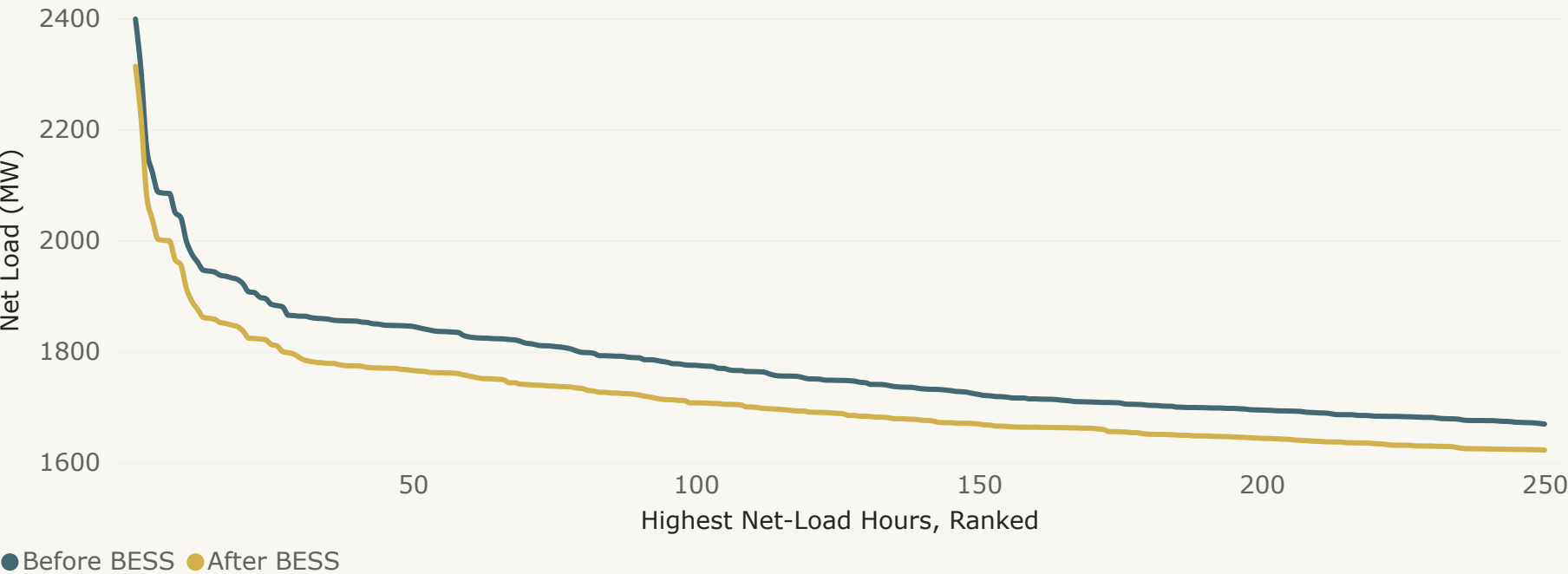
BESS Effect on the Ten Highest Peak-Days

Daily maximum net load before and after storage dispatch



Effect of BESS on High Net-Load Hours

Highest hourly net loads independently ranked before and after BESS dispatch



Interpretation: The duration curve indicates whether storage reduces only the single annual peak or affects a broader set of high-load conditions. A sustained separation between the lines indicates broader peak-management value.

Decision Support and Model Boundaries

Interpreting modeled benefits and remaining gaps

Gross Avoided Energy Cost

\$0.80M

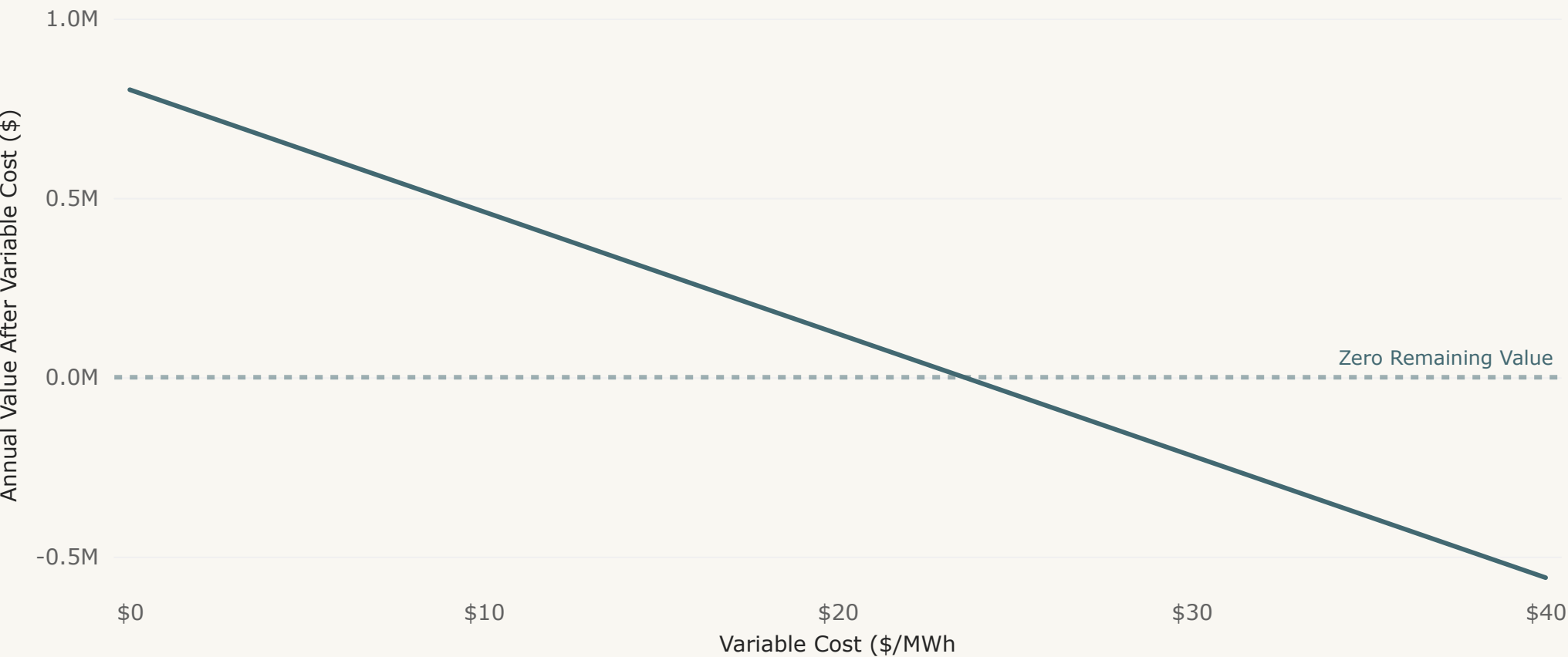
Break-Even Variable Cost

Per MWh discharged

\$23.57

Sensitivity to Battery Variable Cost

Gross avoided cost less assumed degradation and variable operating costs



Decision Conclusion: The modeled BESS demonstrates a credible operational role in winter peak management and selective energy-cost avoidance. The approximately \$0.8 million annual result represents gross avoided marginal energy cost and should not be interpreted as total project benefit or project profitability. The investment case would require additional analysis of capital and lifecycle costs, dependable capacity contribution, ancillary services, network deferral, resilience, and marginal emissions.

Scope of BESS Screening Model

Quantified	Not Quantified
- Hourly battery charging - Hourly battery discharging - Storage energy losses - Change in hourly net load - Annual and monthly peak reduction - Gross avoided marginal energy cost - Operating days and hours - Equivalent full cycles - Seasonal operating patterns - Sensitivity to variable operating cost	- Battery capital cost - Fixed operating cost - Financing and replacement cost - Capacity value - Operating-reserve value - Ancillary-service revenue - Transmission investment deferral - Renewable curtailment recovery - Outage resilience - Marginal emissions reduction - Terminal or residual value

Recommended next steps

1. Test alternative BESS power and duration combinations.
2. Include degradation, fixed operating cost, and replacement assumptions.
3. Estimate dependable capacity contribution during winter peaks.
4. Assess operating-reserve and ancillary-service opportunities.
5. Identify potential transmission or distribution investment deferrals.
6. Obtain marginal-generation and marginal-emissions information.
7. Test additional weather, load-growth, and renewable-development scenarios.
8. Compare lifecycle system benefits against total lifecycle battery cost.